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Agent-based modelling of road traffic congestion at scale

Z.W.T. Mason

Stoch Answers Ltd, Sheffield, UK

Abstract

Three different Agent-based models of traffic flow were evaluated: Nagel-Schreckenberg, constant speed and a multinomial distribution. They were all implemented on an analogue computer. i.e. a Monopoly board.

Subsequent digital computer simulations performed at scale produced data that was analysed. The fundamental diagram for Nagel-Schreckenberg did not appear to relate to real world data whereas the constant speed model matches that of Daganzo. The multinomial model resulted in a smoothed version of the fundamental diagram for constant speed and, due to congestion, the local speeds had a truncated Quartic Exponential distribution. Distributions for local densities and momenta were not categorised.

1. Introduction

Do you want to play a game? Take one Monopoly board and space out the tokens for the players evenly around the 40 squares. Typically, there are 8 tokens so that initially means a gap of 4 squares between tokens. Determine how many squares each player wants to advance and, if necessary to avoid a collision or to stop overtaking, reduce it so that it is equal to the number of squares in the gap to the next token clockwise. Move all the tokens clockwise at the same time and then repeat.

There are various ways to determine how many squares each token would like to move. The classic one being the Nagel-Schreckenberg model [1]. In that case each player needs to take note of the number of squares that they moved last time. If this is less than some maximum value, say 6, then they should add one to the number. Finally roll two dice and, if they both come up even, then take one from the number. Another possibility is to roll two dice and subtract two from their sum. The simplest would be to use a constant value of, say, 10 although some people may find that boring.

It should be noted that periods of boredom are deemed to be good for the development of children's brains. Any questions of "Are we nearly there yet?" can always be answered by a no due to the circuitous route not having a destination. The gaming experience may be enhanced by playing, obviously on repeat, "The Road to Hell" by Chris Rea, a song inspired by driving around the M25 in congestion. This is obviously the perfect holiday game for the traffic engineer and their family however no recommendations are given for divorce lawyers.

The reader may be familiar with what is described above as examples of Agent Based Modelling, a technique which is well known for producing Spherical Cows. Hopefully at least one of the models is at least a qualitative model. To help determine that simulations were performed resulting in averaged data for plotting the fundamental diagram of traffic flow.

Figure 1 shows the fundamental diagram of the Nagel-Schreckenberg model for the choice of track length, speed limit and probability of braking of 40, 6 and 0.3. Figure 2 for a constant speed of 5, the triangular relationship

Email address: `papers@stochanswers.com` (Z.W.T. Mason)

between flow and density being Daganzo's model [2] due to car following. Figure 3 for the sum of two six-sided dice minus two. Of these set of three the Nagel-Schreckenberg model looks the worst although that may be just due to the particular settings.

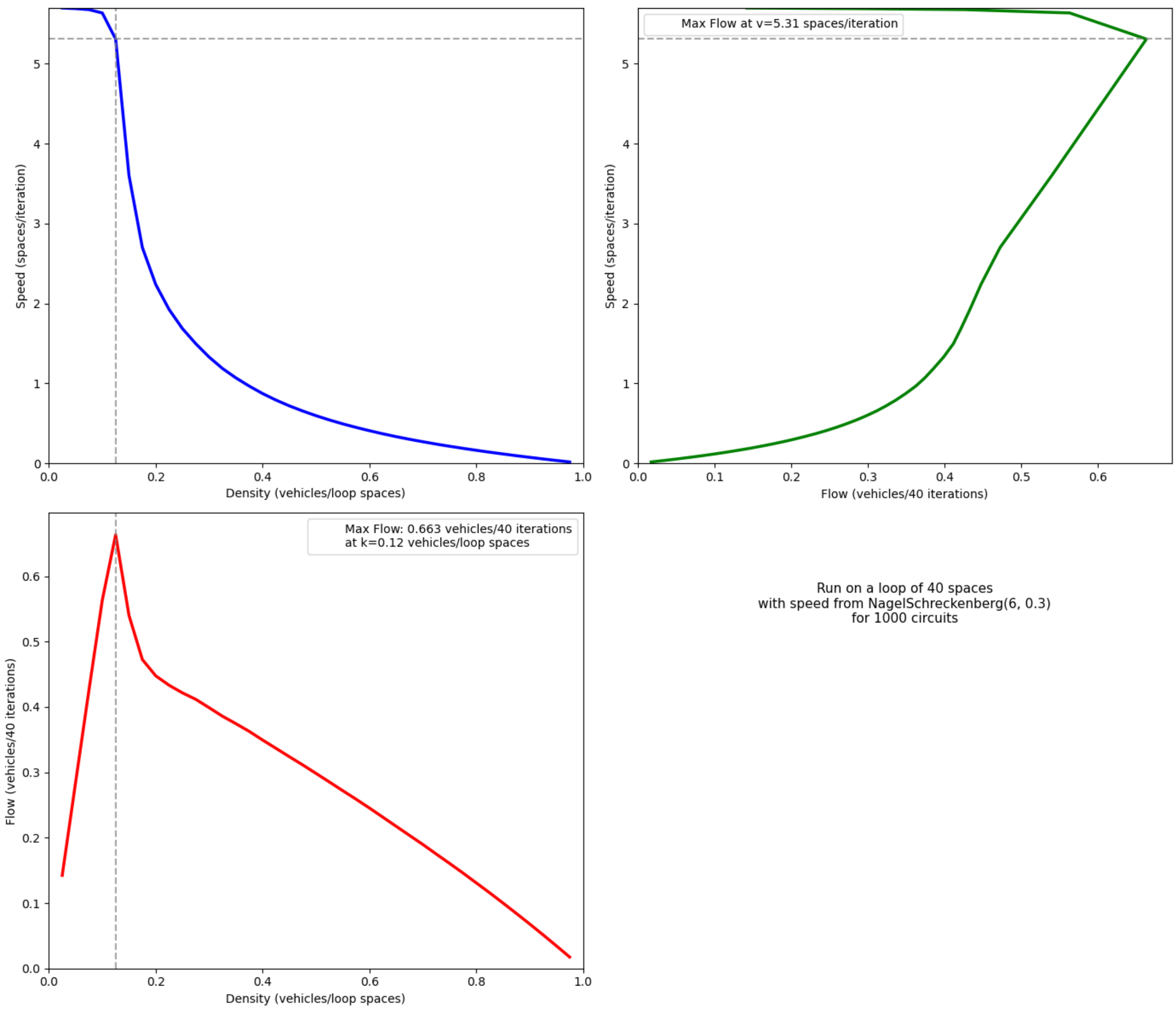

Figure 1. Nagel-Schreckenberg with a speed limit of 6 and a 0.3 probability of braking

2. Probability distributions

Gridlock is easy to model - it is just a Dirac delta distribution with mode at zero speed. The constant speed model produces a bimodal mixture distribution which is the weighted sum of two Dirac delta distributions with modes at gridlock and the constant speed.

The sum of dice is more interesting as it starts with a multinomial distribution which can be approximated by the normal distribution. As the density increases this skews in a way which can possibly be fitted by the generalisation by Fisher [3] of the normal distribution called the Quartic Exponential Distribution (QED). Then the probability distribution becomes truncated to stop the flow going in reverse before converging to the Dirac delta distribution at gridlock. Note that as the variance of a normal distribution goes to zero a Dirac delta is formed.

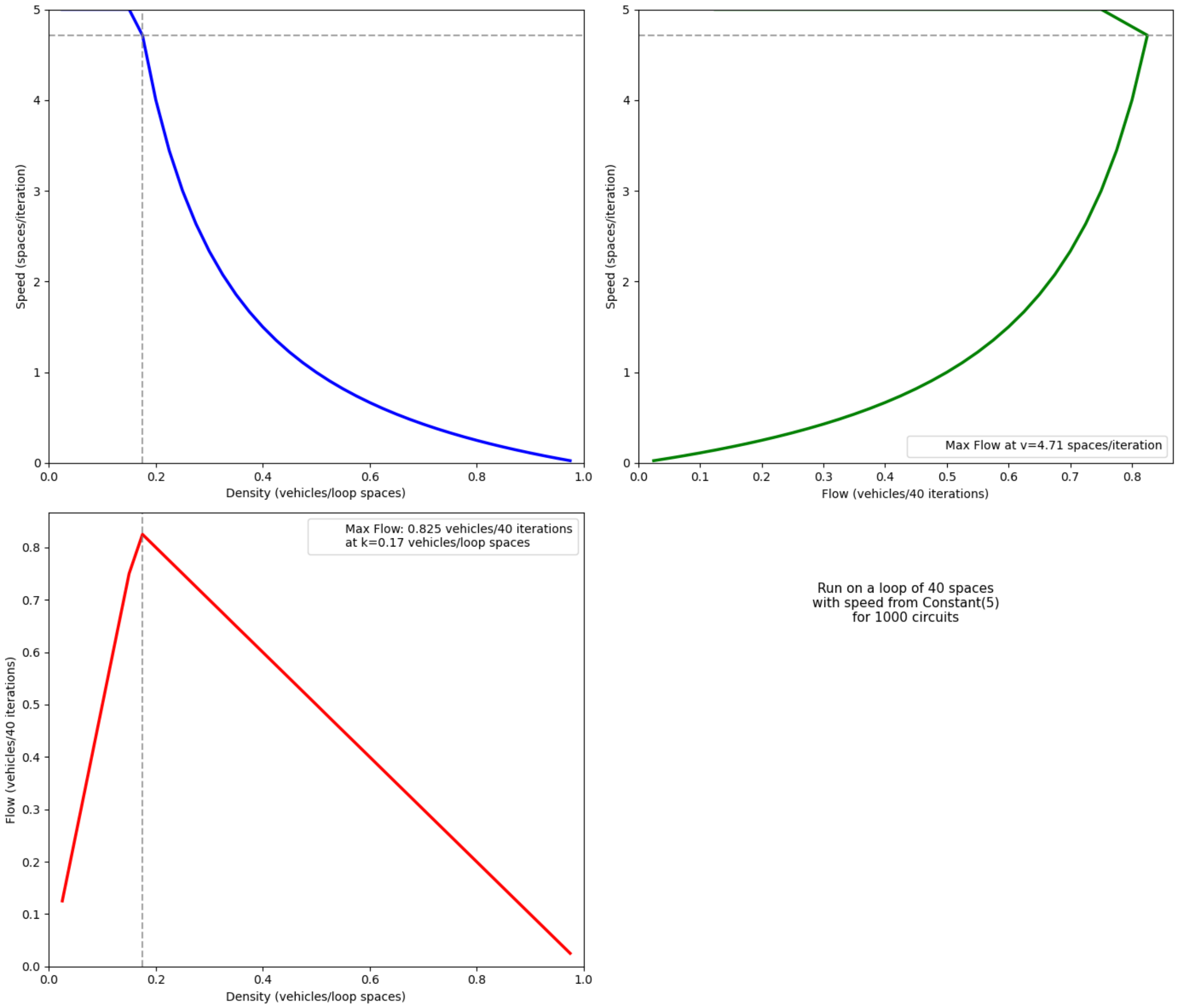

Figure 2. A constant speed of 5

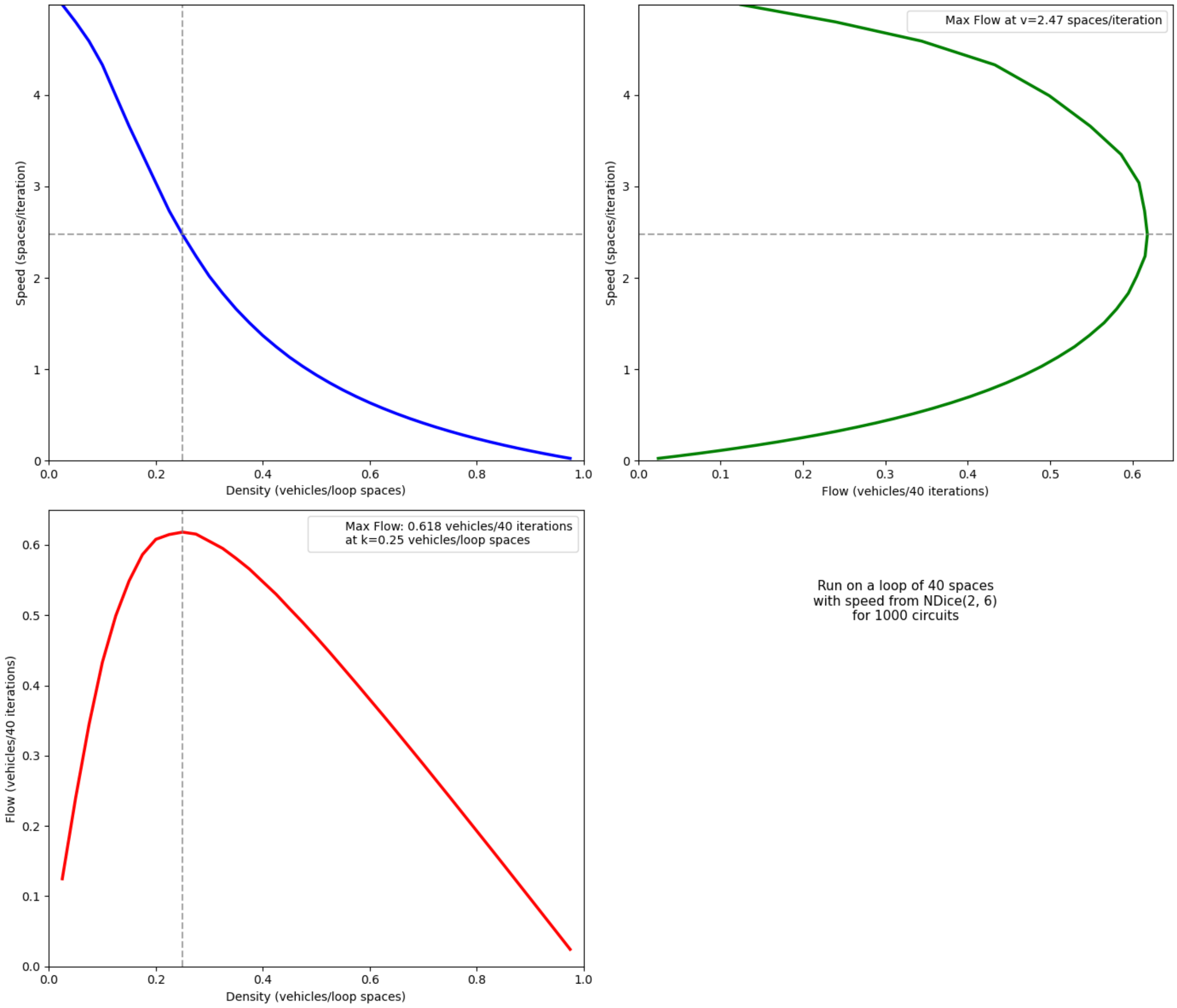

Figure 3. The sum of two six-sided dice minus two

Figure 4 shows the result of fitting a cubic to the derivative of the logarithm of the probability distribution for a track of length 1000 with 100 cars when the rolls of four six-sided dice are summed and then four is subtracted followed by collision avoidance. Note that the probability of being stationary is sufficiently large to indicate a truncated distribution. The skewness is captured by the fit being a cubic rather than the straight line required by the normal distribution.

Denoting the gap to the car in front as f, to the rear as r then for n evenly spaced cars at constant speed on a track of l spaces the density can be defined in two different ways, global and local

$$\frac{n}{l} = \frac{2}{f + r + 2} \tag{1}$$

Figures 5 and 6 are the equivalent histograms for the local densities and momenta to figure 4. Both results have been divided up into 1000 bins between zero and 100% in the first case and zero to maximum speed in the latter.

3. Discussion

It was hoped, at the start of this work, that it would be possible to unify the various ways that traffic is modelled by scaling Agent-based models. Some results of passing interest were generated but nothing fundamental.

Pope [4] noted that one point PDF's cannot capture wave behaviour such as that modelled by Lighthill and Whitham [5] and by Richards [6]. Unfortunately, the discrete results are only continuous at scale and thus the continuum mechanics of Pope's modelling have seemingly no analogue. Pope, however, did form joint PDFs however only the PDF for speed was of sufficient quality.

References

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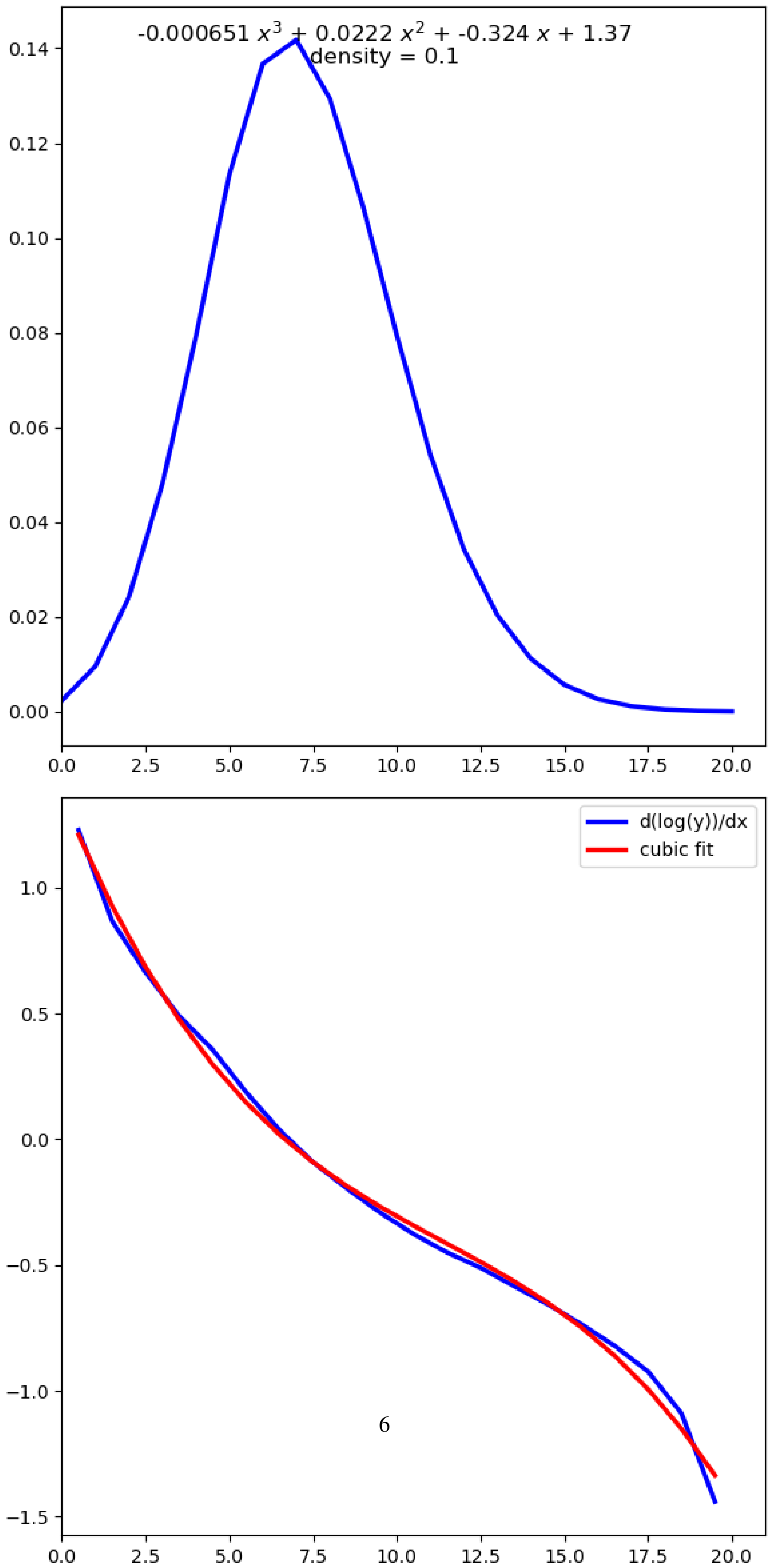

Figure 4. Sum of four dice minus 4 on a track of length 1000 with 100 cars

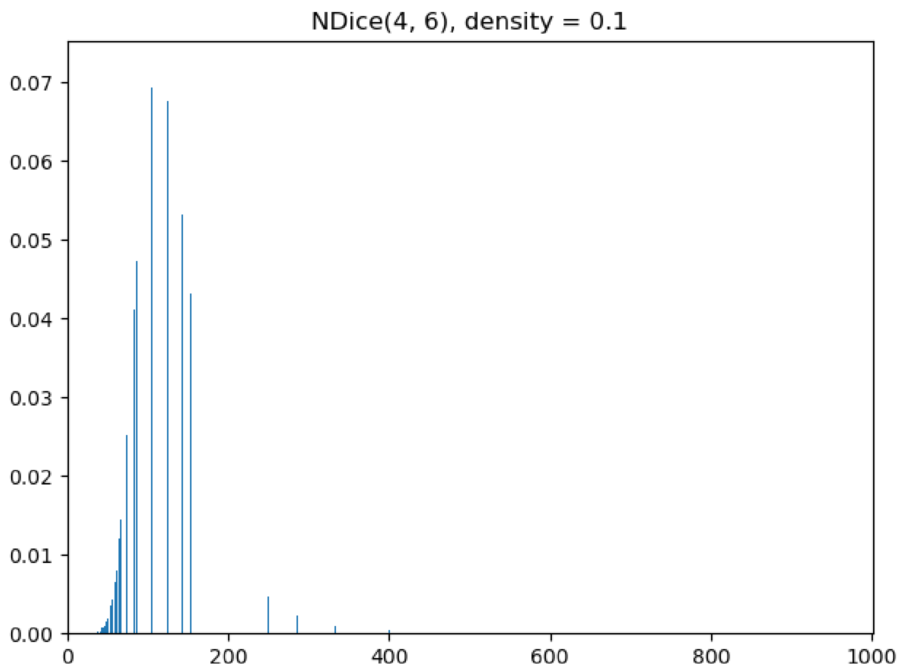

Figure 5. Densities for the sum of four dice minus 4 on a track of length 1000 with 100 cars

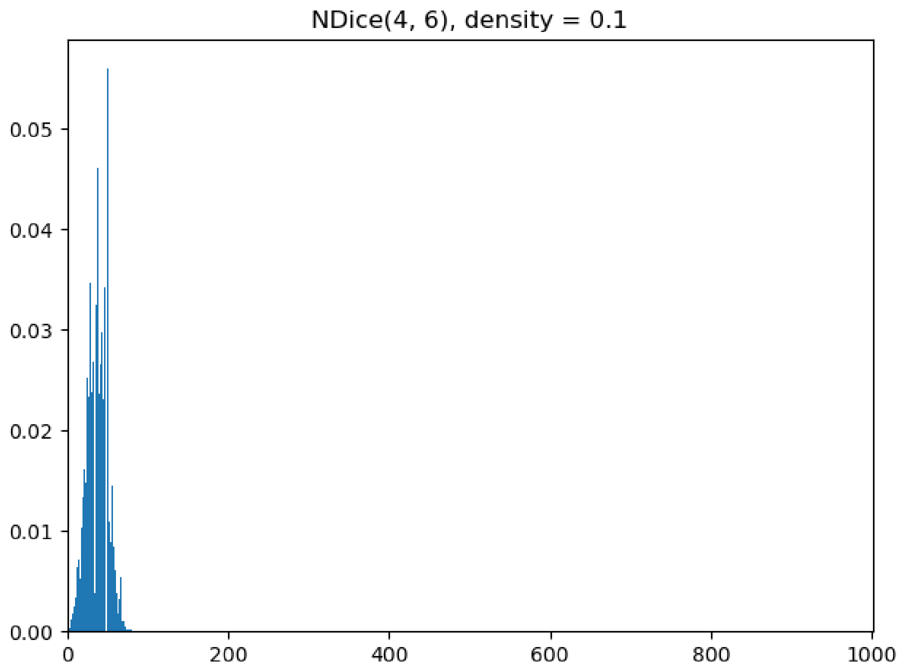

Figure 6. Momenta for the sum of four dice minus 4 on a track of length 1000 with 100 cars